

Unit 14, Lesson 1: Constructing Two-Way Tables



Benchmarks

MA.912.DP.3.1 Construct a two-way frequency table summarizing bivariate categorical data. Interpret joint and marginal frequencies and determine possible associations in terms of a real-world context.

MA.912.DP.1.1 Given a set of data, select an appropriate method to represent the data, depending on whether it is numerical or categorical data and on whether it is univariate or bivariate.

MA.912.DP.1.2 Interpret data distributions represented in various ways. State whether the data is numerical or categorical, whether it is univariate or bivariate and interpret the different components and quantities in the display.



Learning Targets

- ☐ I can construct a two-way frequency table.
- ☐ I can determine joint and marginal frequencies in a two-way frequency table.



14.1.1 Warm-Up: Percent Calculations

1. Newborn babies typically lose some weight in the first week after birth. Weight loss of up to 10% is common. If a newborn baby weighs 8.5 pounds at birth and loses 8% of their weight in the first week of life, how many pounds does the baby lose?
2. Anya earned a score of 75% on her last math quiz. If she earned 18 points, how many total points was the quiz worth?
3. Five hundred gallons of maple sap were boiled to become 12.5 gallons of maple syrup. What percentage of the 500 gallons of sap is the 12.5 gallons of maple syrup?



14.1.2 Guided Instruction: Two-Way Frequency Tables

A 2018 Pew Research Center survey of American teenagers was conducted to analyze trends in social media and technology.

741 teens who use social media were asked what effect they believe social media has had on people their age (mostly positive, mostly negative, neutral).

1. The effect of social media is a ☐ quantitative variable.
☐ categorical variable.

The data when asked what effect they believe social media has had on people their age (mostly positive, mostly negative, neutral) is summarized.

- 219 said the effect was mostly positive
- 211 said the effect was mostly negative
- 311 said the effect was neutral

On the same survey, the Pew Research Center also collected data from teens on how much time they believe they spend with their friends in person (too much, too little, or about the right amount).

2. Time spent with friends in person (too much, too little, or just the right amount) is a ☐ quantitative variable.
☐ categorical variable.

A **two-way frequency table** is a display used to summarize bivariate categorical data. The table can also be called a contingency table or a joint frequency table.

The two-way frequency table displays the results for teens when asked about the effect of social media, in addition to the results when being asked about time spent with friends in person.

Teen Social Media Survey

		Effect on People Your Age			
Time Spent With friends in Person		Mostly Positive	Mostly Negative	Neutral	Total
	Too Much	22	21	20	63
	Too Little	87	77	105	269
	Just Right	110	113	186	409
	Total	219	211	311	741

3. Circle the frequency for those who said the effect of social media was mostly positive.
4. Circle the frequency for those who said the effect of social media was mostly negative.
5. Circle the frequency for those who said the effect of social media was neutral.

6. How many teens reported spending too much time with friends in person? Circle the value in the table.
7. How many teens reported spending too little time with friends in person? Circle the value in the table.
8. How many teens reported spending just the right amount of time with friends in person? Circle the value in the table.

Each of the values circled is a **marginal frequency**.

The total frequency for one category of a categorical variable, which is the total frequency for any row or column in a two-way frequency table, is called a **marginal frequency**.

9. How many people reported that the effect of social media is mostly positive **and** reported that they spend too much time with friends in person?
10. How many people reported that the effect of social media is mostly negative **and** reported that they spend too much time with friends in person?
11. How many people reported that the effect of social media is mostly negative **and** reported that they spend too little time with friends in person?



In a two-way table, a **joint frequency** is the number of times a combination of two conditions occurs.

12. Draw a box around each joint frequency in the table.

13. What percentage of respondents reported that the effect of social media is mostly positive **and** reported that they spend too much time with friends in person?

$$\frac{\boxed{}}{\boxed{}} =$$

14. What percentage of respondents reported the effect of social media is mostly negative **and** reported that they spend too much time with friends in person?

$$\frac{\boxed{}}{\boxed{}} =$$



In a two-way table, a **joint relative frequency** is the ratio of the joint frequency and the total number of data points.



14.1.3 Collaboration: Interpreting a Two-Way Table

- With your partner, use the two-way frequency table to determine if each statement in the table is true or false. If a statement is false, correct the statement to make it true.

Teen Social Media Survey

		Effect on People Your Age			
Time Spent With Friends in Person		Mostly Positive	Mostly Negative	Neutral	Total
	Too Much	22	21	20	63
	Too Little	87	77	105	269
	Just Right	110	113	186	409
	Total	219	211	311	741

269 participants reported that they spend too little time with friends in person.	☐ True ☐ False
More teens reported that they spend too little time with friends than just the right amount with friends in person.	☐ True ☐ False
211 participants reported the effect of social media was mostly negative and spend just the right amount of time with friends in person.	☐ True ☐ False
186 participants reported that social media has a neutral effect and they spend just the right amount of time with their friends.	☐ True ☐ False
There was a total of 1,823 participants in the survey.	☐ True ☐ False



14.1.4 Guided Practice: Constructing a Two-Way Table

- Kieran decides to gather data on hair and eye color for his genetics project in science. Kieran's data is shown.

The table shows the number of people who match each description.

Brown hair, Brown Eyes	Brown Hair, Blue Eyes	Blonde Hair, Brown Eyes	Blonde Hair, Blue Eyes
33	18	11	24

- Construct a two-way frequency table to summarize the data.

			Total
Total			

- b. The total number of students surveyed with brown eyes is _____ .

This value represents a

- ☐ marginal frequency.
- ☐ joint frequency.

- c. The total number of students surveyed with brown eyes and blonde hair is _____ .

This value represents a

- ☐ marginal frequency.
- ☐ joint frequency.



14.1.5 Try It: Constructing a Two-Way Table

1. A survey of high school students asked participants their gender and if they play video games. The data is summarized:
- 68 participants were males who play video games
 - 60 participants were males who do not play video games
 - 145 participants do not play video games
 - There were 279 participants in the survey

- a. Complete the joint frequency table.

Student Video Game Activity

	Play Video Games	Do Not Play Video Games	Total
Male			
Female			
Total			

- b. How many female students and do not play video games?
- c. Of the students who play video games, how many were female?
- d. How many students surveyed play video games?

2. Jackson County Schools, located in Marianna, Florida, conducted a student survey to find out how students spend the majority of their free time outside of school.

The data of this survey is shown in tables below.

Elementary Students		
On the Phone	Playing Video Games	Reading
389	863	1,242

Middle School Students		
On the Phone	Playing Video Games	Reading
532	557	481

High School Students		
On the Phone	Playing Video Games	Reading
978	844	224

Construct a two-way frequency table that shows all of the survey data.



14.1.6 Wrap-Up: Reflection

1. Rate your confidence in creating two-way tables to display bivariate categorical data.



2. What are some questions you still have about two-way frequency tables?